

Prakash Chand Palsaniya

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OBJECTIVE

Backend and AI-focused developer with strong fundamentals in OOP and data structures, and hands-on experience building AI-native applications using LLM APIs, RAG pipelines, and prompt & context engineering, with a first-principles approach to shipping software with AI as a core part of the engineering workflow.

SKILLS

Languages: C++, C, JavaScript (ES6+), SQL

AI / LLM: RAG, LLM APIs (Gemini, OpenRouter), LangChain, LangGraph, Vector DB (Qdrant), Prompt & Context Engineering, SSE Streaming

Backend: Node.js, Express.js, REST API, Microservices, API Gateway, Socket.IO, RabbitMQ, Docker

Databases: MongoDB, PostgreSQL, Redis (Caching)

Tools & DevOps: Git, AWS (EC2, S3, Lambda, CloudFront), Docker Compose, Postman

Core CS: DSA, OOP, DBMS, OS, SDLC

EDUCATION

National Institute of Technology, Srinagar

B.Tech. in Information Technology (CGPA: 8.4)

Srinagar, India

Aug. 2023 – Aug. 2027

PROJECTS

DocuMind | *Agentic RAG chat-with-PDF platform*

[Live Demo](#) | [GitHub](#)

- Built an **agentic RAG** pipeline with **LangGraph** orchestrating multiple nodes (rewrite → retrieve → grade → generate) with a **Tavily web-search** fallback when document context is insufficient.
- Engineered **hybrid retrieval** over a **Qdrant** vector DB — fusing dense embeddings and **BM25** sparse search via Reciprocal Rank Fusion, then a **cross-encoder reranker** for better grounding.
- Designed an async ingestion pipeline with **BullMQ** + **Redis** workers for PDF parsing, chunking, and embedding, decoupling heavy processing from the API for responsive uploads.
- Implemented **SSE token streaming** and a **semantic cache** to serve repeated queries instantly; integrated **Clerk** auth and **AWS S3** storage across decoupled services.
- **Tech Stack:** Node.js, LangGraph, Qdrant, BullMQ, Redis, PostgreSQL, AWS S3, Clerk.

FinMate | *AI-powered personal finance assistant*

[Live Demo](#) | [GitHub](#)

- Engineered an **AI finance assistant** using the **Gemini LLM API** with intent classification to route messages to DB lookups or the AI coach, with a layered **fallback** (LLM → regex → static) for reliability.
- Implemented **real-time token streaming** via SSE, using context engineering that injects recent conversation history for accurate follow-up handling.
- Integrated a **Telegram Bot** via webhook for natural-language transaction parsing; applied Redis caching for aggregation queries and repeated LLM responses, cutting latency by **85%** (800ms → 120ms).
- **Tech Stack:** Node.js, Express.js, MongoDB, Redis, Gemini AI, OpenRouter, AWS EC2.

PKLinks | *Distributed microservices URL shortener*

[Live Demo](#) | [GitHub](#)

- Designed a distributed **6-service microservices** backend with an **API Gateway** routing traffic to Auth, Link, Redirect, and Analytics services.
- Engineered TTL-based Redis caching to optimize performance under high concurrency, cutting latency by **60%**.
- Implemented async analytics via **RabbitMQ** — the Redirect service fires events consumed by an Analytics Worker for **GeoIP & device** parsing.
- Built a complete **Auth Service** with JWT registration, login, and secure password-reset flows; decoupled email delivery via RabbitMQ to a Notification Service.
- **Tech Stack:** Node.js, Express.js, Redis, RabbitMQ, Docker Compose, API Gateway, MongoDB.

ACHIEVEMENTS

Problem Solving: Solved **500+ Data Structures & Algorithms problems** on LeetCode and GeeksforGeeks.

Certifications: Postman API Fundamentals Student Expert – proficiency in building, testing, and documenting RESTful APIs.